📄 Project Documentation – eAuto (Bike Spare Parts E-Commerce Platform)

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1. Aim of the Project

The aim of this project is to **design and develop an e-commerce platform for bike spare parts**, offering a **complete online solution** for both customers and administrators. The system is built to **overcome the drawbacks of traditional offline spare part purchases**, by ensuring a **modern, user-friendly, responsive, and secure platform**.

This project will serve as a **bridge between customers and vendors**, making it easier to purchase spare parts online while helping administrators manage their business efficiently.

Key Objectives

1. User-Friendly Online Shopping Experience

- To enable users to **browse, search, and filter** bike spare parts by **brand, type, and compatibility**.
- To provide **detailed product information**, including specifications, pricing, discounts, stock status, and images.
- To ensure a **responsive design** that works seamlessly across **desktops, tablets, and smartphones**.

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2. Shopping Cart & Checkout System

- To implement a **dynamic cart system** where users can **add, update, or remove products** in real-time.
- To integrate a **secure checkout process** that records delivery address, billing information, and payment details.
- To display **total amount**, including shipping charges, taxes, and discounts before confirming the order.
- To generate an **invoice/receipt** for user reference after successful checkout.

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3. Order Management & Tracking

- To allow customers to **track their orders** from placement to delivery with live status updates.
- To generate **unique order IDs** for reference, customer support, and tracking.

- To maintain a **complete order history** in the customer's profile for easy **reordering or past reference**.

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4. Admin Panel for Management

- To develop a **centralized admin dashboard** for managing products, categories, stock, and pricing.
- To allow admins to **process orders** (update status as pending, shipped, or delivered).
- To provide **user management features**, enabling admins to view registered customers and their activity.
- To generate **reports & analytics** (sales performance, product popularity, user activity).

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5. Database & Backend Integration

- To design a **centralized MySQL database** for handling **users, products, categories, carts, and orders**.
- To ensure **data integrity and security** through constraints, relationships, and validations.
- To implement **authentication and session management** for secure user and admin access.
- To store all transactions in a **consistent and reliable manner** with proper indexing for performance.

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6. Additional Features

- **Advanced search functionality** with filters such as **price, brand, and product type**.
- **Responsive & interactive UI** using **modern web technologies** (React.js, Bootstrap, AOS animations).
- **Scalable design** to support future enhancements (new brands, payment gateways, shipping partners).
- **Reports & analytics** for admins, including **sales reports, product performance insights, and customer activity logs**.

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2. Features of the Project

The proposed **E-Commerce Platform for Bike Spare Parts (eAuto)** is designed to be **efficient, user-friendly, secure, and scalable**.
The system caters to the needs of both **end-users (customers)** and **administrators (admins)** by integrating features that simplify online shopping and streamline management tasks.

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1. User Registration and Login System

- New users can **create an account** by providing personal details such as **name, email, phone number, and password**.
- **Registered users** can securely log in using their email and password.
- **Password encryption** is implemented to ensure sensitive user data is secure.
- **Session management** ensures users remain logged in until they explicitly log out.
- Users can:
    - Update their **profile details**,
    - Change/reset **passwords**,
    - Access their **order history**.

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2. Search and Filter for Bike Spare Parts

- Users can **search products** using keywords (e.g., "Honda Silencer").
- Advanced **filters** allow narrowing down results by:
    - Brand,
    - Category/Type,
    - Price Range,
    - Stock Availability.
- **Quick search suggestions** help users find products faster and improve navigation.

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3. Product Listing with Detailed Information

- Products are displayed with complete details:
    - Image,
    - Product Name,
    - Brand,
    - Price (Original & Discounted),
    - Stock Status,
    - Short Description.
- **Responsive product cards** ensure a clean and attractive UI on all devices.
- **Sorting options** available:
    - Price (Low to High / High to Low),
    - Popularity,
    - Newest First.

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4. Cart Management (Add, Update, Remove Items)

- Users can **add products** to their cart from the product list or detail page.
- Option to **update product quantities** or remove items.

- **Dynamic cart total calculation** including discounts and shipping charges.
- Cart contents are **saved in the database**, so they persist even after logout.

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5. Checkout Process with Delivery Address Management

- Users can proceed to checkout with selected items.
- A **delivery form** captures:
  - Name,
  - Address,
  - City,
  - Zip Code,
  - Contact Number.
- Users can **save multiple addresses** and set a **default delivery address**.
- An **order summary** (products, quantity, total price, shipping charges) is displayed before final confirmation.

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6. Order Placement and Tracking (for Users)

- Each order receives a **unique Order ID** for tracking and support purposes.
- Orders go through multiple statuses:
  - Pending, Confirmed, Shipped, Delivered, Cancelled.
- Users can track their orders in the **"My Orders"** section.
- Complete **order history** is stored in the user's profile for reordering or record keeping.

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7. Admin Panel (Back Office Management)

- **Product Management:** Add, edit, update stock, or delete products.
- **Brand & Category Management:** Organize products effectively.
- **Order Management:** View all orders, update statuses (pending → shipped → delivered).
- **User Management:** View registered users and their activities.
- **Reports Dashboard:** View **sales statistics, product performance, and order insights**.
- **Secure login system** ensures only admins can access the backend.

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8. Reports Generation

- **Product Reports:** Stock availability, top-selling products, most-viewed products.
- **Order Reports:** Breakdown of completed, pending, and cancelled orders.
- **User Reports:** List of registered users and their purchase history.

- These reports assist admins in **business analysis and decision-making**.

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9. Responsive UI (Mobile & Desktop Friendly)

- Built using **Bootstrap 5 and custom CSS** for responsiveness.
- Works seamlessly across all screen sizes (desktop, tablet, mobile).
- Ensures a **consistent user experience** on multiple devices.

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10. Database-Driven System (MySQL)

- A **centralized database** stores all data including:
  - Users,
  - Products,
  - Categories,
  - Brands,
  - Cart,
  - Orders,
  - Delivery Addresses.
- **Data integrity, relationships, and constraints** are maintained using MySQL.
- **Backup and recovery support** ensures data security.
- Indexed queries ensure **fast data retrieval and performance**.

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3. E–R Diagram (Entity Relationship Diagram)

The **Entity–Relationship Diagram (ERD)** of the **eAuto – Bike Spare Parts E-Commerce Platform** represents the logical structure of the database.
It defines the entities, their attributes, and the relationships between them, ensuring that data is organized efficiently for smooth system operations.

Entities & Attributes

1. Admins

- **id** (Primary Key) – Unique identifier for each admin.
- **email** – Email address used for admin login.
- **password** – Encrypted password for authentication.
- **created_at** – Timestamp when the admin account was created.

Purpose:
Admins are responsible for managing products, monitoring orders, and overseeing system operations.

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2. Users

- **id** (Primary Key) – Unique identifier for each registered user.
- **fullname** – Full name of the customer.
- **email** – Unique email for login and notifications.
- **contact** – Phone number of the user.
- **birthdate** – Date of birth for profile details.
- **gender** – Gender of the user.
- **password** – Encrypted password for account security.

Purpose:
Represents customers who browse, shop, and place orders for bike spare parts.

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3. Products

- **id** (Primary Key) – Unique identifier for each product.
- **product_name** – Name of the spare part.
- **price** – Selling price after discounts.
- **original_price** – Original price before discount.
- **discount_amount** – Discount applied on the product.
- **image** – Product image for display.
- **brand** – Brand of the spare part.
- **main_brand** – Parent manufacturer or main brand category.
- **rating** – Average customer rating (stars).
- **reviews** – Number of reviews received.
- **stock_status** – Availability status (e.g., In Stock / Out of Stock).

Purpose:
Represents the bike spare parts available for purchase.

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4. Cart

- **id** (Primary Key) – Unique identifier for each cart entry.
- **user_id** (Foreign Key → Users.id) – References the user who owns the cart.
- **product_id** (Foreign Key → Products.id) – References the product added to the cart.
- **product_name** – Name of the selected product.
- **price** – Price of the product at the time of adding to cart.
- **image** – Product image for cart display.
- **brand** – Brand of the product.

- **quantity** – Quantity of the product selected by the user.

Purpose:
Acts as a **temporary storage** for products selected by users before checkout.

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5. Orders

- **id** (Primary Key) – Unique identifier for each order.
- **user_id** (Foreign Key → Users.id) – References the user who placed the order.
- **fullname** – Customer's full name for delivery.
- **email** – Customer's email for notifications.
- **phone** – Customer's contact number.
- **address** – Delivery address.
- **city** – Delivery city.
- **state** – Delivery state.
- **zip** – Postal code.
- **total_amount** – Total cost of the order.
- **payment_method** – Payment type (e.g., Cash on Delivery).
- **created_at** – Timestamp of order placement.

Purpose:
Stores confirmed purchase details after checkout.

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6. Order_Items

- **id** (Primary Key) – Unique identifier for each order item.
- **order_id** (Foreign Key → Orders.id) – References the order this item belongs to.
- **product_id** (Foreign Key → Products.id) – References the purchased product.
- **product_name** – Name of the product ordered.
- **quantity** – Quantity purchased.
- **price** – Price of the product at purchase time.

Purpose:
Stores the list of products inside an order, allowing one order to contain multiple items.

Relationships & Cardinality

- **Admins → Products (1:N)**
  One Admin can manage multiple products (add, edit, delete).
- **Users → Orders (1:N)**
  One User can place multiple orders, but each order belongs to only one user.
- **Orders → Order_Items (1:N)**
  One Order can contain multiple items, but each Order_Item belongs to a single order.

- **Products → Order_Items (1:N)**
  One Product can appear in multiple order items across different orders.
- **Users → Cart (1:N)**
  One User can have multiple items in their cart, but each cart entry belongs to a single user.

Explanation

- **Admins Manage Products:** Admins handle product listings, stock updates, and pricing.
- **Users Browse & Purchase:** Users browse spare parts, add them to cart, and proceed to checkout.
- **Cart as Temporary Storage:** The cart holds user-selected products until order confirmation. Items in the cart are linked to both **Users** and **Products**.
- **Orders & Order_Items:** At checkout, the cart is converted into an order. Each order contains detailed order items (with product, quantity, and price).
- **Product Reusability:** A single product can exist in multiple orders since different users may purchase the same item.
- **Order Management:** Orders are tied to specific users, while Order_Items store detailed product-level breakdown for billing, reports, and tracking.

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4. Data Flow Diagram (DFD)

The **Data Flow Diagram (DFD)** describes how information flows through the **eAuto – E-Commerce Platform for Bike Spare Parts**.
It illustrates the interaction between **external entities** (Users, Admins) and the system, along with the processes and data stores.

Level 0 – Context Diagram

At the **highest level of abstraction**, the entire **eAuto System** is represented as a **single process** that communicates with external entities.

Entities & Interactions

- **User ↔ eAuto System**
  - **Input:** Registration/Login, Search Products, Add to Cart, Place Orders, Track Orders, View Profile.
  - **Output:** User account, product details, updated cart, order confirmations, order tracking, profile info.
- **Admin ↔ eAuto System**
  - **Input:** Add/Edit/Delete Products, Manage Users, Manage Orders, Generate Reports.

- **Output:** Updated product catalog, order updates, user database, and reports for analysis.

 This level shows **what the system does** but not **how it does it**.

Level 1 – Detailed Process View

At Level 1, the **single system process** is broken down into **sub-processes**, showing **how Users and Admins interact with the system** and **how data is processed internally**.

User Activities

1. Register/Login

- **Input:** User details (name, email, password, contact).
- **Process:** Authentication and account creation using database validation.
- **Output:** User profile created; login session established.

2. Browse Products

- **Input:** Search queries, filters (brand, category, price range).
- **Process:** Retrieve product data from **Products Database**.
- **Output:** Product listing with details (name, image, price, discount, stock).

3. Manage Cart

- **Input:** Product selection (add, update quantity, remove).
- **Process:** Store cart items linked to specific user in **Cart Database**.
- **Output:** Updated cart summary with total price displayed to user.

4. Place Orders

- **Input:** Delivery details (address, city, state, zip, phone), payment method, selected cart items.
- **Process:** Convert cart items into an **Order**, store in **Orders & Order_Items Database**.
- **Output:** Order confirmation with unique **Order ID**.

5. Track Orders

- **Input:** User selects from their order history.
- **Process:** Fetch order and order_items details from database.
- **Output:** Order status (Pending, Shipped, Delivered, Cancelled) with tracking info.

Admin Activities

1. Manage Products

- **Input:** Product details (name, price, stock, brand, image).
- **Process:** Add, update, or delete product records in **Products Database**.
- **Output:** Updated product listing visible to users.

2. Manage Users

- **Input:** User account information.
- **Process:** View, update, or deactivate user accounts.
- **Output:** Updated **Users Database**.

3. Manage Orders

- **Input:** Orders placed by customers.
- **Process:** Update order statuses (Pending → Shipped → Delivered/Cancelled).
- **Output:** Updated order tracking visible to users.

4. Generate Reports

- **Input:** Database records (products, users, orders).
- **Process:** Summarize and generate business reports (sales, stock, user activity).
- **Output:** Reports for admin analysis and decision-making.

Data Stores in DFD

- **Users Database** → Stores registered user information.
- **Products Database** → Stores all spare parts details.
- **Cart Database** → Stores user-selected cart items.
- **Orders Database** → Stores confirmed orders.
- **Order_Items Database** → Stores order-specific product details.

5. Flow Chart (System Flow)

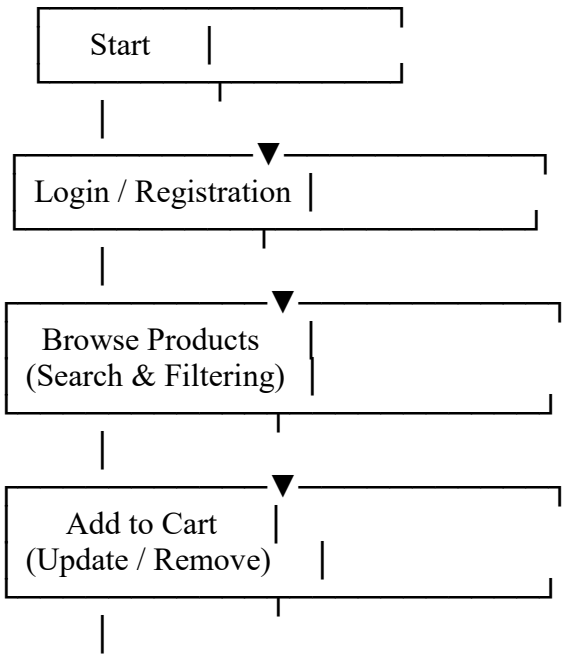
Step–by–Step Flow Description

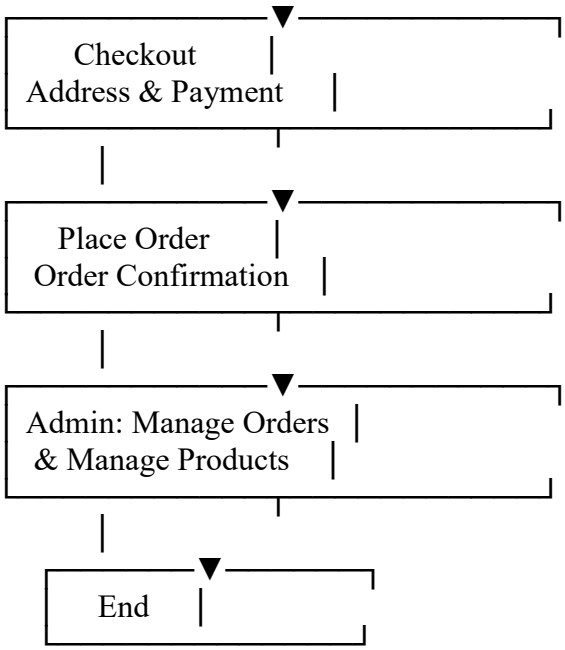
1. **Start**
    - The system begins when the user visits the eAuto platform.
2. **User Login / Registration**
    - If the user is new, they register with details (name, email, password, contact, etc.).
    - If already registered, they log in with email and password.
    - System validates credentials.
    - On success → user dashboard. On failure → error message.
3. **Browse Products → Search & Filter**

- Users can view products listed by brand, category, and price.
- Features: search bar, category filter, sorting (price, popularity, rating).
- Product details include: name, price, discount, brand, stock status, and image.

4. **Add to Cart → Update / Remove**
   - From product page, user can add products to cart.
   - Cart allows **updating quantities**, or **removing items**.
   - Total amount updates dynamically.
5. **Checkout → Enter Address → Payment**
   - User proceeds to checkout.
   - Enters delivery address (name, address, city, state, zip, phone).
   - Selects payment method (Cash on Delivery / Online Payment).
   - System validates details.
6. **Place Order → Order Confirmation**
   - Once order is placed:
     - System generates **Order ID**.
     - Stores order details in the database.
     - Sends confirmation message/email to the user.
     - Updates stock in products.
7. **Admin → Manage Products & Orders**
   - Admin logs into the system.
   - Can add/update/delete products (CRUD).
   - Can view/manage all user orders (approve, ship, cancel).
   - Can generate sales and stock reports.
8. **End**
   - User completes purchase.
   - Admin ensures smooth processing.
   - The process ends until the next interaction.

Flow Chart (Visual Representation)





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6. Limitations of the Project

1. **No Integration with Online Payment Gateways (Cash on Delivery Only)**
   - The current system only supports Cash on Delivery (COD) as the payment method.
   - Popular online payment options such as credit/debit cards, UPI, net banking, or wallets (e.g., Google Pay, Paytm) are not integrated.
   - This limits convenience for customers who prefer digital payments and may reduce the potential user base.
2. **Limited to Bike Spare Parts Domain (Not a Generic E-Commerce Platform)**
   - The platform is specifically designed for **bike spare parts only**.
   - Unlike large e-commerce systems (e.g., Amazon, Flipkart), it cannot handle multiple industries or varied product categories.
   - The database schema, product catalog, and features are domain-specific, which restricts flexibility.
3. **Basic User Interface (UI) with Scope for Enhancement**
   - The UI is functional but **not highly modern or interactive**.
   - It lacks advanced design features such as smooth animations, dark mode, or fully customizable themes.
   - User experience (UX) can be further improved with features like live product previews, auto-suggestions, and enhanced responsiveness.
4. **Reporting is Basic; Advanced Analytics Not Included**
   - Reports are limited to **basic data summaries** such as product lists, user details, and order histories.
   - Advanced analytics features (e.g., sales forecasting, revenue trends, customer purchase behavior analysis) are not available.
   - No integration with visualization dashboards or third-party analytics tools.
5. **No AI-Based Recommendation System (Future Enhancement Possible)**
   - The system does not include **personalized recommendations** such as “Frequently Bought Together” or “You May Also Like.”

- AI/ML-based algorithms that analyze user behavior and preferences are not implemented.
- This reduces the opportunity to boost sales through cross-selling and up-selling strategies.
